TITLE: A STUDY TO DETERMINE THE RELATIONSHIP BETWEEN THE USAGE OF ONLINE ACADEMIC RESOURCES AND ACADEMIC PERFORMANCE

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INTRODUCTION

The rapid advancement of digital technology has significantly transformed the way students access educational materials and engage with learning. Online educational resources such as video lectures on platforms like YouTube, digital notes and e-books, artificial intelligence tools such as ChatGPT, and online learning platforms like Coursera and Udemy have become widely used by college students. These resources allow students to access information easily, review difficult concepts, and learn beyond the limitations of the traditional classroom environment. As a result, online learning tools have become an important supplement to formal education.

The increasing availability of online educational resources has generated interest in understanding their impact on students' academic performance. Online learning platforms can provide flexibility, personalized learning opportunities, and additional academic support for students (Means et al., 2014). Digital resources may help students improve their conceptual understanding by providing multiple explanations and interactive learning materials. However, excessive, or unstructured use of online resources may also lead to distractions or reduced engagement with traditional study methods.

Previous research suggests that the integration of digital technology in education can positively influence student learning outcomes when used effectively (Selwyn, 2016). The accessibility of online resources has made it possible for students to supplement classroom learning with additional explanations, tutorials, and practice materials. At the same time, the effectiveness of these tools often depends on how students use them and the amount of time they devote to such resources.

This study aims to examine the relationship between the use of online educational resources and academic performance among college students. The research focuses on students from five colleges located in the Bhawanipur area of Kolkata, namely Acharya Jagadish Chandra Bose College, The Bhawanipur Education Society College, Gokhale Memorial Girls' College, Asutosh College, and Shri Shikshayatan College. Students from different academic streams were included in the survey. In this study, the number of hours spent using online educational resources serves as the independent variable, while students' Semester Grade Point Average (SGPA) represents the dependent variable. The study seeks to determine whether the use of online educational resources has a measurable relationship with students' academic performance.

OBJECTIVES

The main objective of this study is to examine the relationship between the usage of online educational resources and the academic performance of college students. The specific objectives of the study are as follows:

1. To examine the extent of usage of online educational resources such as YouTube, ChatGPT, PDFs, online books, and online learning platforms among college students.
2. To analyse the relationship between the number of hours spent using online educational resources and students' academic performance measured through Semester Grade Point Average (SGPA).
3. To evaluate whether the use of online educational resources contributes positively to students' academic performance.

METHODOLOGY

Research Design

This study adopts a quantitative research design to examine the relationship between the usage of online educational resources and students' academic performance. The analysis is based on primary data collected through a structured questionnaire. Regression analysis is used as the primary statistical method to determine whether a relationship exists between the hours spent using online educational resources and students' Semester Grade Point Average (SGPA).

Population and Study Area

The study was conducted in the **Bhawanipur area of Kolkata**, focusing on students enrolled in five colleges: Acharya Jagadish Chandra Bose College, The Bhawanipur Education Society College, Gokhale Memorial Girls' College, Asutosh College, and Shri Shikshayatan College. Students from all academic streams were considered for the study.

The total student population across the selected colleges was **24,323 students**.

Sampling Technique

The study employed **proportionate stratified sampling** to ensure fair representation of students from each college. In this method, the population is divided into distinct groups or strata based on a common characteristic, which in this case is the college attended (Creswell & Creswell, 2018). Each college represents a separate stratum.

The number of responses selected from each college was determined using the following formula:

Sample from each college = [(Number of students from each college)/Total number of students] *40

Sample Size

A total of **40 student responses** were collected for the study. The proportional distribution of the sample across the selected colleges is shown below:

College	Total Students	Calculated Sample	Final Sample
Acharya Jagadish Chandra Bose College	4185	6.88	7
The Bhawanipur Education Society College	7488	12.31	12
Gokhale Memorial Girls' College	2350	3.86	4
Asutosh College	6000	9.86	10
Shri Shikshayatan College	4300	7.07	7
TOTAL	**24,323**		**40**

The calculated values were rounded to the nearest whole number to determine the final number of responses from each college.

Data Collection

Primary data for the study was collected through a **Google Form questionnaire** distributed among students from the selected colleges. The questionnaire consisted of ten questions designed to gather information on students' usage of online educational resources and their academic performance.

Variables Used in the Study

The study examines the relationship between the following variables:

Hours spent using online educational resources such as YouTube, ChatGPT, digital PDFs, online books, and online learning platforms. **This is the Independent Variable (X).**

Students' academic performance measured through **Semester Grade Point Average (SGPA)**. **This is the Dependent Variable (Y).**

Data Preparation and Analysis

The collected data was organized and processed using **Microsoft Excel**. Regression analysis was conducted using the Data Analysis tool in Excel to examine the relationship between online educational resource usage and academic performance.

The regression model used in the study is expressed as:

$Y = a+bX$

where, Y = Academic Performance (SGPA)

X = Hours spent using online educational resources

a = Intercept

b = Regression Coefficient

The significance of the regression results was evaluated using **p-values and hypothesis testing** to determine whether the relationship between the variables was statistically significant.

Hypothesis Testing

To determine whether a statistically significant relationship exists between the variables, the following hypotheses were formulated:

Null Hypothesis (H_0): There is no significant relationship between the usage of online educational resources and students' academic performance.

Alternative Hypothesis (H_1): There is a significant relationship between the usage of online educational resources and students' academic performance.

A **significance level of 0.05** was used to test the hypothesis.

PRESENTATION OF DATA

A) DESCRIPTIVE STATISTICS

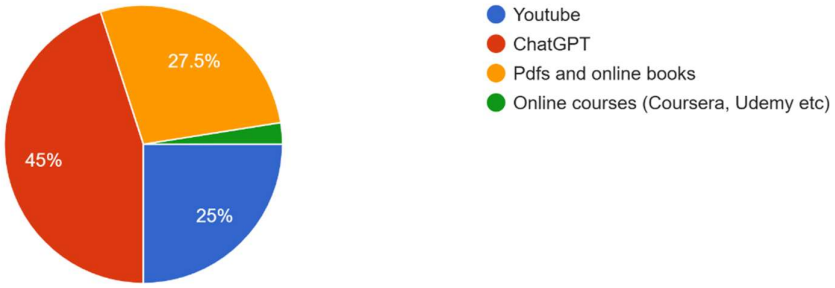
Descriptive statistics were calculated to summarize the key characteristics of the data collected from the respondents. The analysis focuses on the two primary variables used in the study: the **number of hours** students spend using online educational resources and their Semester Grade Point Average (**SGPA).** Other than that, the monthly expenditure on such resources and the number of days of the week, such resources were used, are also considered as other variables**.**

VARIABLES	**MEAN**	**MEDIAN**	**MINIMUM VALUES**	**MAXIMUM VALUES**
Hours spent using online academic resources	3.9875	3.75	0.5	9
The SGPA of the students	7.37	7.365	6	9
Days of the week	5.4	6	1	7
Monthly Expenditure	92.5	0	0	2000

B) CHARTS

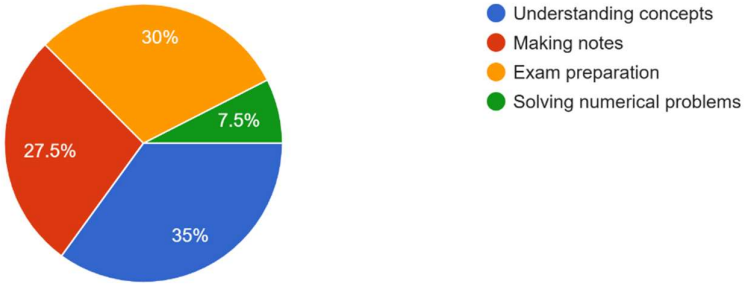
Which online resources do you use the most for studying?

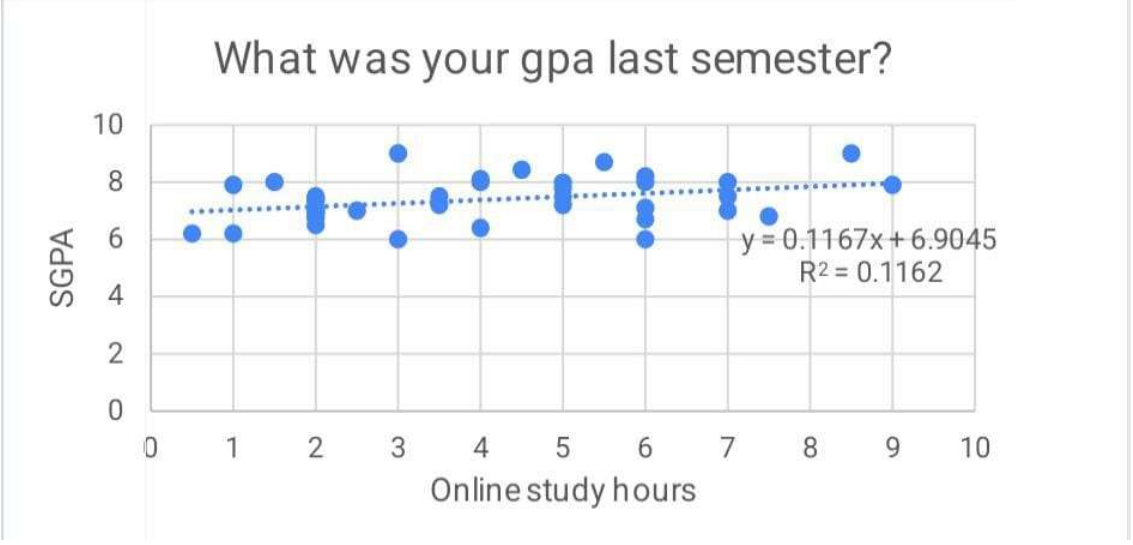
40 responses



What is the purpose of your use?

40 responses





INTERPRETATION AND ANALYSIS

The regression results indicate a **positive relationship** between online study hours and academic performance. The estimated regression equation is:

Academic Performance (SGPA) = 6.904 + 0.116 × (Online Study Hours)

This coefficient suggests that, on average, an additional hour spent using online educational resources is associated with an increase of approximately 0.116 points in SGPA.

The overall regression model was found to be **statistically significant**, $p = 0.031$. Since the **p-value is less than the significance level of 0.05**, the null hypothesis stating that there is no relationship between online study hours and academic performance is rejected. This indicates that the number of hours students spend using online educational resources has a statistically significant effect on their academic performance.

The coefficient of determination (R^2) is 0.116, which indicates that approximately 11.6% of the variation in students' academic performance can be explained by the number of hours spent using online educational resources. While this shows that online study hours do have a measurable influence on academic performance, the relatively low R^2 value suggests that other factors such as study habits, classroom learning, motivation, and access to learning materials may also play an important role in determining students' academic outcomes.

Overall, the results suggest that online educational resources can positively contribute to students' academic performance, although they represent only one of several factors affecting academic achievement.

RECOMMENDATIONS

Based on the findings of the study, several recommendations can be proposed to improve the effective use of online educational resources and enhance students' academic performance.

First, students should be encouraged to use online educational resources in a structured and purposeful manner. Platforms such as YouTube, digital academic materials, online books, and online learning platforms can provide valuable supplementary learning content. When used effectively, these resources can help students clarify concepts, revise course material, and gain additional explanations beyond classroom teaching.

Second, educational institutions can play an important role by guiding students toward reliable and academically relevant online platforms. Colleges may recommend credible educational channels, digital libraries, and structured online courses that complement classroom instruction and support independent learning.

Third, students should maintain a balanced approach between online learning and traditional study methods. Although online educational resources can positively influence academic performance, they should not replace textbooks, lectures, and classroom discussions. A combination of both digital and traditional learning methods can help students achieve better academic outcomes.

Finally, future research could expand on this study by including larger sample sizes and additional variables, such as study habits, classroom attendance, motivation, and access to learning materials. Including these factors may provide a more comprehensive understanding of the determinants of academic performance among college students.

CONCLUSION

This study examined the relationship between the use of online educational resources and the academic performance of college students in the Bhawanipur area of Kolkata. Using primary data collected through a questionnaire and analysed via simple linear regression, the study found a **positive and statistically significant relationship** between online study hours and students' Semester Grade Point Average (SGPA).

The regression results indicate that increased time spent using online educational resources is associated with a modest increase in academic performance. However, the **R^2 value (0.116)** suggests that online study hours explain only a small portion of the variation in SGPA, implying that academic outcomes are also influenced by other factors such as study habits, classroom engagement, and individual learning strategies.

Overall, the findings highlight that online educational resources can serve as valuable supplementary tools in supporting students' learning when used effectively alongside traditional methods of study. As digital learning continues to expand in higher education, understanding how students integrate these resources into their study practices remains an important area for further research.

Regression Study: Online Educational Resource Usage and Academic Performance

This study is purely for educational purposes and all the answers will remain confidential.

~~* Indicates required question~~

1. What is your gender? *

Check all that apply.

- [ ] Male
- [ ] Female
- [ ] Other

2. Which college are you from? *

_____

3. How many hours per day do you use online resources for studying ? *

_____

4. How many days of the week do you use online resource for studies? *

_____

5. Which online resources do you use the most for studying? *

Mark only one oval.

- ◯ Youtube
- ◯ ChatGPT
- ◯ Pdfs and online books
- ◯ Online courses (Coursera, Udemy etc)

6. What is the purpose of your use? *

Mark only one oval.

- ◯ Understanding concepts
- ◯ Making notes
- ◯ Exam preparation
- ◯ Solving numerical problems

7. Do you think online resources help you understand your subject better than textbooks? *

Check all that apply.

- ☐ Yes
- ☐ No
- ☐ Maybe

8. What problems do you face while using online resources? *

Check all that apply.

- ☐ Distracting
- ☐ Information overload
- ☐ It's not always free
- ☐ No major problems

9. What was your gpa last semester? *

_____

10. Monthly expenditure on online educational resources (if any) *

_____

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